

1. Creating and Renaming Files/Directories

Create a directory named `test_dir` using `mkdir`.

Inside `test_dir`, create an empty file called `example.txt`.

Rename `example.txt` to `renamed_example.txt` using `mv`

Answer:

'mkdir' is the command used to create new directory/folder in the given name

'ls' used to list all the files and folders in the current directory

'cd' will change directory to the mentioned path

'touch' will create a new file using the given name in the current directory

'mv' will help to move the file to a specific directory or rename the files.

```
Select rahavi@DESKTOP-TIDIB06: /mnt/e/My Learnings/LinuxLearnings/assignment-1/test_dir
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1$ mkdir test_dir
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1$ ls
test_dir
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1$ cd test_dir
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1/test_dir$ touch example.txt
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1/test_dir$ ls
example.txt
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1/test_dir$ mv example.txt renamed_example.txt
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1/test_dir$ ls
renamed_example.txt
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1/test_dir$
```

2. Viewing File Contents

Use `cat` to display the contents of `/etc/passwd`.

Display only the first 5 lines of `/etc/passwd` using `head`.

Display only the last 5 lines of `/etc/passwd` using `tail`.

Answer:

'cat' will display the contents of the file mentioned.

‘**head**’ is used to display only the top few lines or mentioned number of lines from top.

‘**tail**’ is used to display only the bottom few lines or the mentioned number of lines from the bottom.

```
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1/test_dir
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1/test_dir$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mail List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534:./nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:./usr/sbin/nologin
dhcpcd:x:996:996:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
messagebus:x:995:995:System Message Bus:/nonexistent:/usr/sbin/nologin
syslog:x:100:101:./nonexistent:/usr/sbin/nologin
systemd-resolve:x:989:989:systemd Resolver:./usr/sbin/nologin
chrony:x:988:988:Chrony Daemon:/var/lib/chrony:/usr/sbin/nologin
landscape:x:101:103:./var/lib/landscape:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:./usr/sbin/nologin
rahavi:x:1000:1000:./home/rahavi:/bin/bash
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1/test_dir$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1/test_dir$ tail -n 5 /etc/passwd
systemd-resolve:x:989:989:systemd Resolver:./usr/sbin/nologin
chrony:x:988:988:Chrony Daemon:/var/lib/chrony:/usr/sbin/nologin
landscape:x:101:103:./var/lib/landscape:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:./usr/sbin/nologin
rahavi:x:1000:1000:./home/rahavi:/bin/bash
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1/test_dir$
```

3.Searching for Patterns

Use grep to find all lines containing the word "root" in /etc/passwd.

Answer:

‘**grep**’ is the command used to search for the mentioned string. It is usually combined with the ‘**cat**’ command.

Adding ‘**-n**’ will display the number of times the searched word is present.

```
rahavi@DESKTOP-TIDIBO6: /mnt/e/My Learnings/LinuxLearnings/assignment-1/test_dir
rahavi@DESKTOP-TIDIBO6:/mnt/e/My Learnings/LinuxLearnings/assignment-1/test_dir$ cat /etc/passwd | grep root
root:x:0:0:root:/root:/bin/bash
rahavi@DESKTOP-TIDIBO6:/mnt/e/My Learnings/LinuxLearnings/assignment-1/test_dir$ cat /etc/passwd | grep -n root
1:root:x:0:0:root:/root:/bin/bash
rahavi@DESKTOP-TIDIBO6:/mnt/e/My Learnings/LinuxLearnings/assignment-1/test_dir$ _
```

4. Zipping and Unzipping

Compress the test_dir directory into a file named test_dir.zip using zip.

Unzip test_dir.zip into a new directory named unzipped_dir.

Answer:

'Zip' is the command used to zip the folders in the mentioned name

Ex: zip zipped_dir.zip test_dir

Here zipped_dir is the name in which the folder is going to be zipped and saved
test-dir is the file to be zipped.

This will only zip the outer folder not the files inside of it. If we want to zip everything inside the folder then we need to use **'-r'** to recursively zip all the files inside the mentioned folder.

Ex: zip -r zipped_dir.zip test_dir

'unzip' is the command used to unzip the mentioned folder which will extract the folder and its contents onto the current directory.

Ex: unzip zipped_dir.zip

If we want to unzip/extract the folder into a specific directory name, then we need to mention destination **'-d'**

Ex: unzip zipped_dir.zip -d unzipped_dir

```

Select rahavi@DESKTOP-TIDIB06: /mnt/e/My Learnings/LinuxLearnings/assignment-1
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1$ zip test_dir.zip test_dir
Command 'zip' not found, but can be installed with:
sudo apt install zip
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1$ sudo apt install zip
[sudo: authenticate] Password:
Installing:
  zip
Installing dependencies:
  unzip
Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 0
  Download size: 355 kB
  Space needed: 988 kB / 1025 GB available
Continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 unzip amd64 6.0-29ubuntu1 [180 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 zip amd64 3.0-15ubuntu3 [175 kB]
Fetched 355 kB in 0s (725 kB/s)
Selecting previously unselected package unzip.
(Reading database ... 35923 files and directories currently installed.)
Preparing to unpack .../unzip_6.0-29ubuntu1_amd64.deb ...
Unpacking unzip (6.0-29ubuntu1) ...
Selecting previously unselected package zip.
Preparing to unpack .../zip_3.0-15ubuntu3_amd64.deb ...
Unpacking zip (3.0-15ubuntu3) ...
Setting up unzip (6.0-29ubuntu1) ...
Setting up zip (3.0-15ubuntu3) ...
Processing triggers for man-db (2.13.1-1build1) ...
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1$ zip test_dir.zip test_dir
  adding: test_dir/ (stored 0%)
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1$ unzip test_dir.zip -d unzipped_dir
Archive: test_dir.zip
  creating: unzipped_dir/test_dir/
  chmod (directory attributes) error: Operation not permitted
warning: cannot set modif./access times for unzipped_dir/test_dir/
  Operation not permitted
warning: cannot set permissions for unzipped_dir/test_dir/
  Operation not permitted
warning: set times/attrs failed for unzipped_dir/test_dir/
  failed setting times/attrs for 1 dir entries
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1$ ls
'Screenshot (1).png'  'Screenshot (2).png'  'Screenshot (3).png'
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1$ zip -r test_dir.zip test_dir
  adding: test_dir/ (stored 0%)
  adding: test_dir/renamed_example.txt (stored 0%)
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1$ unzip test_dir.zip -d unzipped_dir
Archive: test_dir.zip
  creating: unzipped_dir/test_dir/
  chmod (directory attributes) error: Operation not permitted
  extracting: unzipped_dir/test_dir/renamed_example.txt  #chmod (file attributes) error: Operation not permitted
  (warning) cannot set modif./access times
  Operation not permitted
warning: cannot set modif./access times for unzipped_dir/test_dir/
  Operation not permitted
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1$

```

5. Downloading Files

Use `wget` to download a file from a URL (e.g., <https://example.com/sample.txt>).

Answer:

‘**wget**’ is the command to download the http request response in the current directory.

```

rahavi@DESKTOP-TIDIB06: ~
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1$
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1$
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1$
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1$ wget https://example.com/sample.txt
--2026-05-08 19:29:22--  https://example.com/sample.txt
Resolving example.com (example.com)... 172.66.147.243, 104.20.23.154, 2606:4700:10::6814:179a, ...
Connecting to example.com (example.com)|172.66.147.243|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-05-08 19:29:22 ERROR 404: Not Found.

```

6. Changing Permissions

Create a file named `secure.txt` and change its permissions to read-only for everyone using `chmod`.

Answer:

'**sudo**' is the command used to give a user administrator access.

'**chmod**' is the command used to change the read-write-execute access of a directory/file.

4 represents the read access 2 represents the write access and 1 represents the execute access.

3 digits represents Owner, Group and other permissions

444 will give read-only access to everyone.

```
rahavi@DESKTOP-TIDIB06: ~  
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1$ cd test_dir  
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1/test_dir$ touch secure.txt  
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1/test_dir$ chmod 444 secure.txt  
chmod: Operation not permitted (os error 1)  
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1/test_dir$ sudo chmod 444 secure.txt  
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1/test_dir$ ls -l  
total 0  
-rwxrwxrwx 1 root root 0 May  8 19:13 renamed_example.txt  
-r-xr-xr-x 1 root root 0 May  8 19:32 secure.txt  
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1/test_dir$ mv secure.txt ~/  
rahavi@DESKTOP-TIDIB06:/mnt/e/My Learnings/LinuxLearnings/assignment-1/test_dir$ cd ~  
rahavi@DESKTOP-TIDIB06:~$ ls  
secure.txt  
rahavi@DESKTOP-TIDIB06:~$ chmod 444 secure.txt  
rahavi@DESKTOP-TIDIB06:~$ ls -l  
total 0  
-r--r--r-- 1 rahavi rahavi 0 May  8 19:32 secure.txt  
rahavi@DESKTOP-TIDIB06:~$
```

7. Working with Environment Variables

Use export to set a new environment variable called MY_VAR with the value "Hello, Linux!".

Answer:

‘export’ is the command used to set a new environment variable.

‘printenv’ is the command used to print all the environment variables set in the current distro.

```
rahavi@DESKTOP-TIDIB06:~  
rahavi@DESKTOP-TIDIB06:~$ export MY_VAR="Hello, Linux!"  
rahavi@DESKTOP-TIDIB06:~$ printenv  
SHELL=/bin/bash  
MSL2_GUI_APPS_ENABLED=1  
PS1_DISTRO_NAME=ubuntu  
NAME=DESKTOP-TIDIB06  
PWD=/home/rahavi  
LOGNAME=rahavi  
HOME=/home/rahavi  
LANG=C.UTF-8  
MSL_INTEROP=/run/MSL/A37_interop  
MY_VAR=Hello, Linux!  
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lзма=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bzip=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.taz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.xz=01;31:*.cab=01;31:*.wim=01;31:*.sum=01;31:*.dmg=01;31:*.swf=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xpm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.png=01;35:*.mov=01;35:*.mpe=01;35:*.mpeg=01;35:*.mp4=01;35:*.mp4v=01;35:*.m4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*.n=00;90:*.n=00;90:*.bak=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.swp=00;90:*.tmp=00;90:*.dpkg-dist=00;90:*.dpkg-old=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsave=00;90:  
WAYLAND_DISPLAY=wayland-0  
LESSCLOSE=/usr/bin/lesspipe %s %s  
TERM=xterm-256color  
LESSOPEN=| /usr/bin/lesspipe %s  
USER=rahavi  
DISPLAY=:0  
SHLVL=1  
XDG_RUNTIME_DIR=/run/user/1000  
MSLENV=  
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/ao/lib:/mnt/c/Program Files (x86)/Microsoft SDKs/Azure/CLI2/bin:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS/mn  
t/c/WINDOWS/system32/wbem:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/c/Users/Admin/AppData/Roaming/npm:/mnt/c/Program Files/nodejs:/mnt/c/ProgramData/chocolatey/bin:/mn  
t/c/Program Files/Git/cmd:/mnt/c/Users/Admin/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/Admin/AppData/Roaming/npm:/mnt/c/Users/Admin/AppData/Roaming/nvm:/mnt/c/Program Files/nodejs:/mnt/c/Users/Admin/AppDa  
ta/Local/GitHubDesktop/bin:/mnt/c/Users/Admin/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/Admin/AppData/Local/Programs/Azure Dev CLI:/mnt/c/Users/Admin/AppData/Local/Programs/cursor/resources/app/  
bin  
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus  
HOSTTYPE=x86_64  
PULSE_SERVER=unix:/mnt/ssl/PulseServer  
OLDPWD=/mnt/c/My Learnings/LinuxLearnings/assignment-1/test_dir  
_/usr/bin/printenv  
rahavi@DESKTOP-TIDIB06:~$
```

Submission Guidelines -: Attach Screenshots or command along with explanation and submit in doc(google doc or microsoft doc) format also attach github repo link.